cases these should be equal partnerships, not the United States insisting on calling the shots. For the robotic exploration of Mars, such programs are already under way. For human flight, the chief such activity is clearly the international space station. Eventually, we might muster joint simulated planetary missions in low Earth orbit. One of the principal objectives of these programs should be to build a tradition of cooperative technical excellence.

- Technological development—using state-of-the-art robotics and artificial intelligence—of rovers, balloons, and aircraft for the exploration of Mars, and implementation of the first international return sample mission. Robotic spacecraft that can return samples from Mars can be tested on near-Earth asteroids and the Moon. Samples returned from carefully selected regions of the Moon can have their ages determined and contribute in a fundamental way to our understanding of the early history of the Earth.
- Further development of technologies to manufacture fuel and oxidizer out of Martian materials. In one estimate, based on a prototype instrument designed by Robert Zubrin and colleagues at the Martin Marietta Corporation, several kilograms of Martian soil can be automatically returned to Earth using a modest and reliable Delta launch vehicle, all for no more than a song (comparatively speaking).
- Simulations on Earth of long-duration trips to Mars, concentrating on potential social and psychological problems.
- Vigorous pursuit of new technologies such as constant-thrust propulsion to get us to Mars quickly; this may be essential if the radiation or microgravity hazards make one-year (or longer) flight times too risky.
- Intensive study of near-Earth asteroids, which may provide superior intermediate-timescale objectives for human exploration than does the Moon.
- A greater emphasis on science—including the fundamental sciences behind space exploration, and the thorough analysis of data already obtained—by NASA and other space agencies.